Group projects Financial Modeling I

Deadline: 30/04/2025, 23:59, please submit your *.py or *.ipynb files and a pdf file with the answers using the designated folder on the K2 site of the course.

Project 1: Group 6 (Maxime Yilmaz, Alexandre Paquet, Christian Cheng), Group 7 (Zuer Li,Qiuyan Li,LUONG Thi Lan Hue), Group 5 (Célia Dompiétrini, Loane Mboulou, Zaineb Erraji), Group 11 (Anas BENELGHAZI, Luc GAULTIER), Group 12 (Pierre ACHARD, Enzo BEN HASSINE,Meygan FAGUNDES), Group 22 (Dronova Polina, Damian Loustau, Girardi Luca Matteo)

Linear regression for independetly identically sampled data

The task is to predict the logarithm of the house prices (variable $lprice$) using the following variables: logarithm of the lot size (variable $llotisize$), logarithm of the surface of the house (variabl $lsqrtft$), number of bedrooms (variable $bdrms$). The .csv file is called hprice1_merged.csv.

1. Choose randomly two sets of 40 data points from your data set, and for each data set compute the mean, variance, and the histogram of each of the variables $lprice$, $llotsize$, $lsqrtft$, $bdrms$. Are these values similar for the two data sets ? Can we say that each of these variables are independetly identically sampled ? Consider using hypothesis testing to compare the distribution of these two data sets.

2. Estimate the coefficients of the linear regression from data. Reserve at least 20% of your data for validation.

3. Calculate the R^2 score and the average mean squared error. Comment on the quality of the model.

4. Evalute the confidence intervals for the estimated regression parameters. Are all regression coefficients statistically significant ? Explain your answer.

5. Add the variable $\log(lotsize^2 * sqrft)$ as a new variable, and construct a linear regression model predicting $lprice$ using $llotsize$ and $lsqrtft$ and $bdrms$ and $\log(lotsize^2 * sqrft)$. What can you say about the relevance of $llotsize$ and $lsqrtft$ and $bdrms$ for predicting $lprice$? Is it consistent with the results of the previous model (the one without $\log(lotsize^2 * sqrft)$). Explain your answer. Is there a linear relationship among those variables ?

Time-series prediction

Use the data set fertil31 .csv

to predict fertility rate (variable gfr) using the fertility rate of the previous years. Use linear regression. Consider detrending using quadratic dependence on time. Evaluate the R^2 score and try taking into account several past values: fertility rate of the previous year, of the previous two years, etc.

Project 2: Group 8 (Killian Bruni,Maïlys Texier), Group 9 (ZHANG Ruge,CAO Ranjie), Group 10 (Ngoc Qutnh PHAM, Thanushree Anilkumar, Pierre-Henri Jean André COUFFON DE TREVROS), Group 13 (Yixin Gao,Qianlei Weng, Xue Lu), Group 23 (Chengjie He, Hu Xin, Klouz Mehdi)

Linear regression for independetly identically sampled data

The task is to predict house prices in different communities based on air pollution level (represented by the concentration of nitrogine oxid), distance from employment centers, average student-teacher ratio in local schools and average number of rooms, and crime rate The variables are as follows:

- *price*
  median housing price
- *crime*
  crimes committed per capita
- *nox*
  nitrogine oxide, parts per 100 mill.
- *rooms*
  avg number of rooms per house

- *dist*
  weighted dist. to 5 employ centers

- *stratio*
  average student-teacher ratio

- *lprice*
  $\log(price)$

- *lnox*
  $\log(nox)$

The .csv file is hprice2_merged.csv.

- Choose randomly two sets of 100 data points from your data set, and for each data set compute the mean, variance, and the histogram of each of the variables *lprice*, *lnox*, *rooms*, *dist*. Are these values similar for the two data sets ? Can we say that each of these variables are independetly identically sampled ? Consider using hypothesis testing to compare the distribution of these two data sets.

- Find a linear regression model which uses the variables *lnox*, *dist*, *rooms*, *stratio* in order to predict the variable *lprice*. When estimating the coefficients of the linear regression, reserve at least 20% of your data for validation.

- Calculate the R^2 score. Comment on the quality of the model. Are all the independent variables also linearly independent ?

- Evalute the confidence intervals for the estimated regression parameters. Are all regression coefficients statistically significant ? Explain your answer.

- Find a linear regression model which uses the variables *lnox*, *dist*, *rooms*, *stratio* and the new variable $\log(nox^3 * e^{dist+1})$ to predict *lprice*. Compute the confidence intervals of the estimated regression parameters. What can you say about the significance of the variables *lnox*, *dist*, *rooms*, *stratio* in predicting *lprice*. Is it different from the significance in the first model ? If so, why ? Is there a linear relationship among the variables ?

Time-series prediction

Use the data set perturbed_priceindex.csv to predict the consumer price index based on its values in the previous years. Use linear regression. Consider detrending using quadratic dependence on time. Evaluate the R^2 score and try taking into account several past values: price index of the previous month, of the previous two months, etc.

Project 4: Group 1 (Lana Szymoniak,Florine Rousteau), Group 3 (Vincent Malgrange,Victor Bouffant,Léo Rioufrays–Laplace), Group 15 (HE Chengjie,ZHOU Haipeng, QU Siyi), Group 26 (Fadi Ballegi, Jody Aria Widjaya, Rajneesh Rajbhar, Justin Durand), Group 19 (Marlowe Venier-Falk, Mehul Nagori, Choraria Himanshu), Group 24 (Calixte Wesley,Yasmine Zemzemi, Haipeng Zhou)

Linear regression for independetly identically sampled data

The task to predict logarithm of wages based on education, experience and IQ, hours worked. The variables are as follows:

- *lwage*
  logarithm monthly earnings
- *hours*
  average weekly hours
- *IQ*
  IQ score
- *educ*
  years of education
- *exper*
  years of work experience

The .csv file is Wage2_merged.csv.

1. Choose randomly two sets of 100 data points from your data set, and for each data set compute the mean, variance, and the histogram of each of the variables *lwage*, *hours*, *IQ*, *educ*, *exper*. Are these values similar for the two data sets ? Can we say that each of these variables are independetly identically sampled ? Consider using hypothesis testing to compare the distribution of these two data sets.

2. Find a linear regression model for predicting *lwage* using the variables $\log(hours + 1)$, $\log(exper + 1)$ and *educ*. Leave at least 20% of the data for validation.

3. Calculate the R^2 score. Comment on the quality of the model.

4. Evalute the confidence intervals for the estimated regression parameters. Are all regression coefficients statistically significant ? Explain your answer.

5. Build a linear regression model for predicting *lwage* using the variables $\log(hours+1)$, $\log(exper+1)$ and *educ* and IQ and $\log((exper+1)*\sqrt{(hours+1)})$. Compute the confidence intervals of the estimated regression parameters. What can you say about the significance of the variables $\log(hours+1)$, $\log(exper+1)$ and *educ* and in predicting *lwage*. Is it different from the significance in the first model ? If so, why ? Is there a linear relationship among the variables ?

Time-series prediction

Use the data set from pcip_filtered .csv to predict the percentage change in monthly industrial production with respect to the previous year (variable *pcip*). Use linear regression. Consider detrending using affine dependence on time. Evaluate the R^2 score and try taking into account several past values: the value of *pcip* the past year, the year before, etc.

Project 4: Group 2 (Thelma GRECO,Angelick ORANGE), Group 5 (Yipan Liu,Ruixuan Li,Jingyan Wu), Group 16 (DAMI Tommaso, CASELLI Clara, FAUROUX Hannah), Group 17 (DULON Bettina, DAHAN Julie), Group 20 (Emanuele Giuseppe Scichilone, Natalija Vukotic, Ahimin Amethier Martin Paul), Group 25 (Benjelloun Lila,Chanelle Stark, Agha Baihass Sabet)

Linear regression for independetly identically sampled data

The task is to predict the percentage of students receiving a passing grade in math (variable $math10$) in function of the following variables:

- $lnchprg$ percentage of students getting lunch at school
- $staff$ number of teachers per 1000 students
- $expend$ expenditure per student
- $salary$ average teacher salary
- $graduation$ graduation rate (percentage)
- $benefits$ average benefits (outside salary) of teachers.

The .csv file is meap93.csv.

1. Choose randomly two sets of 100 data points from your data set, and for each data set compute the mean, variance, and the histogram of each of the variables $lnchprg$, $staff$, $expend$, $salary$, $math10$. Are these values similar for the two data sets ? Can we say that each of these variables are independetly identically sampled ? Consider using hypothesis testing to compare the distribution of these two data sets.

2. Find a linear regression model for predicting $math10$ using the variables $lnchprg$, $\log(expend)$, $\log(salary)$. Find linear regression models for predicting $math10$ using only $lnchprg$, only $\log(expend)$ and only $\log(salary)$. Leave at least 20% of the data for validation.

3. Calculate the R^2 score for each of the 4 models. Comment on the quality of the model.

4. Evalute the confidence intervals for the estimated regression parameters for each of the for 4 models. Are all regression coefficients statistically significant ? Explain your answer.

5. What can you say about the significance of the variables $\log(expend)$, $\log(salary)$ and $lnchprg$ in predicting $math10$. Do the coefficients of $\log(expend)$, $lnchprg$, $\log(salary)$ change a lot when you use all 3 variables to predict $math10$ compared to the case when you use only one of them ? For example, is the coefficient of $\log(expend)$ in the model $\widehat{math10} = \beta_0 + \beta_1 \log(expend)$ is similar to the coefficient of $\log(expend)$ in the model $\widehat{math10} = \beta_0 + \beta_1 \log(expend) + \beta_2 \log(salary) + \beta_3 lnchprg$. If so, why ? Is there a linear relationship among the variables ?

Time-series prediction

Use the data set barium_import_filtered.csv to predict the annual barium import (variable $chnimp$). Use linear regression. Consider detrending using affine dependence on time. Evaluate the R^2 score and try taking into account several past values: the value of $chnimp$ the past year, the year before, etc.